

Supplementary Appendix
Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab
versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV
Melanoma

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1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S9 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ($RFS \leq DMFS$), and a patient cannot die without having both events ($DMFS \leq OS$). The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic

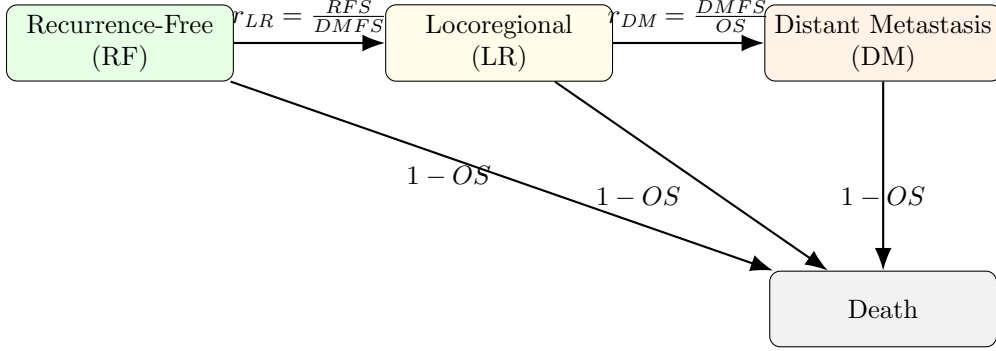


Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7 (<https://automeris.io/WebPlotDigitizer>). Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS (60-month DMFS was not available from the published figure). The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual point extraction with 5–10 points per curve per time point by a single reviewer. The 60-month landmark values (68.8% vs 49.1% for RFS; 92.2% vs 71.3% for OS) match the published landmark rates from the 5-year ASCO 2026 update [4], confirming the accuracy of the digitization process.

Comparison with published landmark rates: The published 5-year landmark rates (RFS: 68.8% vs 49.1%; OS: 92.2% vs 71.3%) are identical to the digitized values, indicating

minimal extraction error. At earlier time points, the published 18-month RFS rates from the primary analysis (79% vs 62%) [5] differ from the digitized values (80.4% vs 62.2%) by 1.4 percentage points (combo) and 0.2 percentage points (pembro), consistent with typical digitization error for KM curves.

Availability of confidence intervals: The 5-year RFS landmark rates were reported with 95% confidence intervals: 68.8% (95% CI 56.3%–78.3%) for the combination arm and 49.1% (95% CI 33.3%–63.0%) for the pembrolizumab arm. These CIs were not incorporated into the primary analysis but are reported here for transparency.

2.2 Observed vs. Fitted Survival Curves

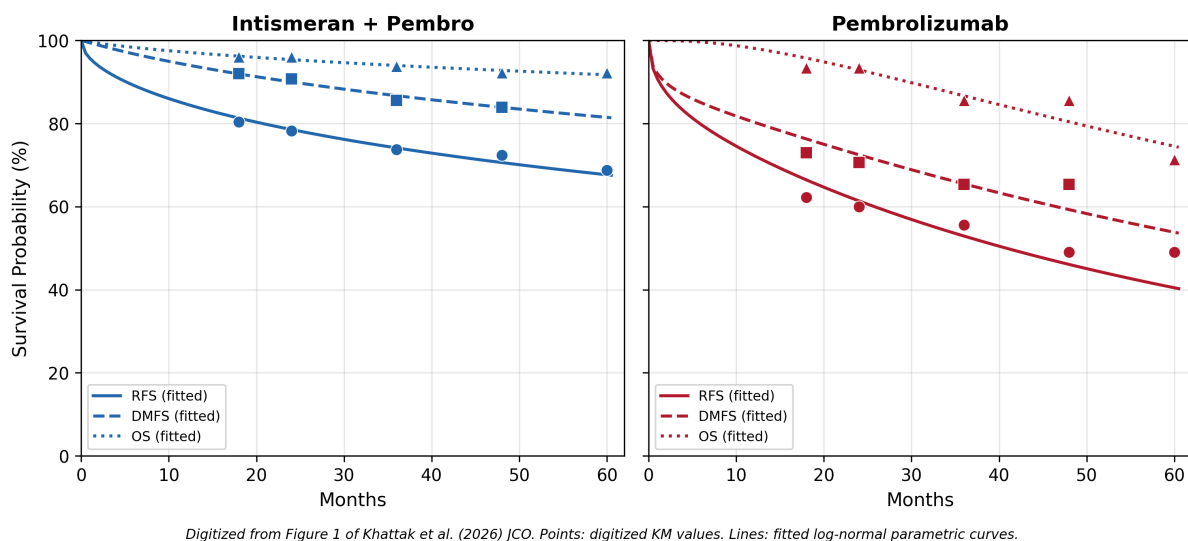


Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm. RFS = recurrence-free survival; DMFS = distant metastasis-free survival; OS = overall survival. Digitized points are extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7.

3 Survival Model Selection

3.1 Component-Level Parametric Fit

Table S3: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

3.2 Joint Model Selection

A total of 4,096 distribution combinations (4 distributions per 6 curves = 4^6) were evaluated for structural validity at 40 years (480 monthly cycles). All 4,096 combinations satisfied the RFS $\leq$ DMFS $\leq$ OS ordering constraint and monotonicity (0 crossing violations). The full results

are available online (`output/survival_combos_4096.csv`).

Table S4: Top 10 Joint Model Combinations by Total AIC

Rank	OS	r _{DM}	r _{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / LL	LL / W	-222.61	76.3%	0.0%
2	LN / W	LN / LL	LL / LL	-222.52	76.3%	0.0%
3	LN / W	Exp / LL	LL / W	-222.47	76.3%	0.0%
4	LLog / W	LN / LL	LL / W	-222.42	72.9%	0.0%
5	LN / W	LN / LL	LN / W	-222.40	76.3%	0.0%
6	LN / W	LN / LL	LL / LN	-222.39	76.3%	0.0%
7	LN / W	Exp / LL	LL / LL	-222.38	76.3%	0.0%
8	W / W	LN / LL	LL / W	-222.37	71.1%	0.0%
9	LLog / W	LN / LL	LL / LL	-222.33	72.9%	0.0%
10	LN / W	LN / LL	LN / LL	-222.31	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential. C/P = Combo/Pembro.

Key finding: While all 4,096 combinations are structurally valid, the long-term OS extrapolation diverges dramatically (Figure S3). The top-ranked combinations all use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3%, which is substantially higher than general population survival at age 101 (approximately 0%). This discrepancy motivated the general population mortality constraint applied in the primary analysis.

3.3 Lifetime OS Extrapolation by Distribution

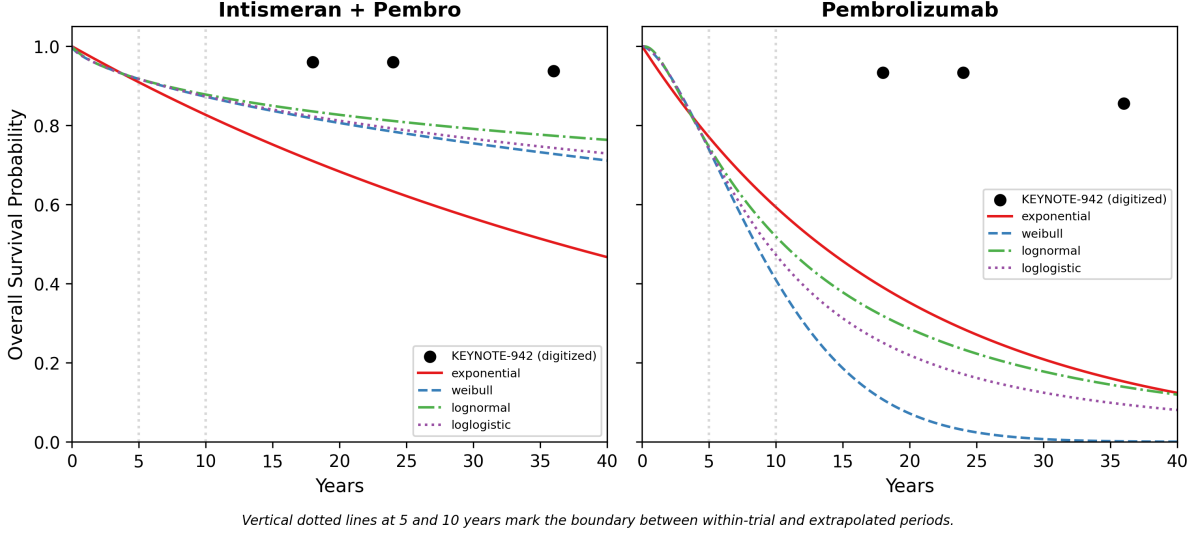


Figure S3: Lifetime overall survival extrapolation by parametric distribution for the intismeran plus pembrolizumab arm (left) and pembrolizumab alone arm (right). Digitized KEYNOTE-942 data points are shown in black. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; the 5–10 year period is between the two dotted lines; beyond 10 years is pure extrapolation. The four distributions fit the within-trial data similarly (RMSE < 0.04 for all), yet produce wildly divergent 40-year survival estimates, from 0% (weibull) to 76.3% (log-normal). This illustrates why model selection cannot rely on within-trial fit alone and must incorporate external validity constraints.

4 General-Population Mortality and Structural Validation

4.1 GP Mortality Implementation

The general-population mortality constraint is applied as a “mortality floor” that caps the modeled overall survival at the age- and sex-matched US general population survival. This prevents the parametric extrapolation from producing implausibly long survival estimates in the long-term tail.

Life table source: US 2019 period life tables (Arias et al., 2022, NVSS).

Baseline age: 61 years (median age of the KEYNOTE-942 trial population).

Sex weighting: 64% male, 36% female, matching the KEYNOTE-942 trial demographic distribution. The life table survival was computed as a weighted average of the male and female survival curves.

Interpolation and monthly conversion: The annual life table survival probabilities were linearly interpolated to monthly resolution. Let $S_{GP}(y)$ be the GP survival at integer year y ; the monthly value at month m is:

$$S_{GP}(m/12) = (1 - \alpha) \cdot S_{GP}(\lfloor m/12 \rfloor) + \alpha \cdot S_{GP}(\lceil m/12 \rceil) \quad (1)$$

where $\alpha = m/12 - \lfloor m/12 \rfloor$.

Constraint onset: The GP mortality floor is applied from month 60 onward. Before month 60, the observed KEYNOTE-942 survival data dominate and the parametric extrapolation is unconstrained:

$$S_{\text{model}}(t) = \begin{cases} S_{\text{param}}(t), & t < 60 \text{ months} \\ \min[S_{\text{param}}(t), S_{GP}(t)], & t \geq 60 \text{ months} \end{cases} \quad (2)$$

Both arms (intismeran + pembro and pembro alone) are subject to the same background mortality floor.

Motivation: As shown in Appendix 3, the log-normal OS extrapolation for the combo arm yields 40-year survival of 76.3% in the unconstrained model, which is inconsistent with US general population survival at age 101 (approximately 0%). The GP mortality constraint ensures that long-term survival projections are biologically plausible by anchoring the extrapolation to empirical population mortality rates.

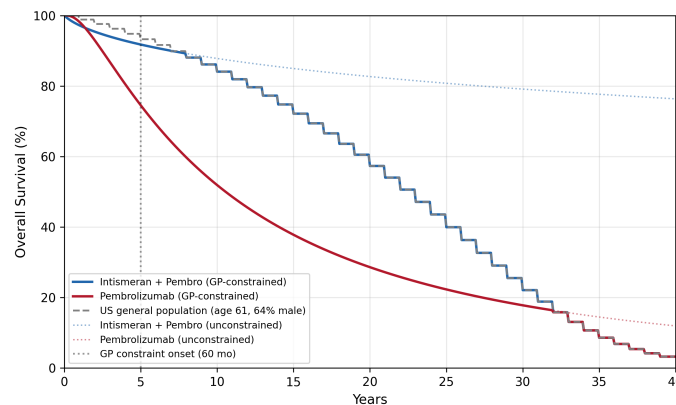
4.2 Long-Term Survival Diagnostics

Table S5: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival	Constrained?
5	91.8%	74.5%	93.3%	No (pre-constraint)
10	84.1%	51.9%	84.1%	Yes (combo at GP)
15	72.1%	37.8%	72.1%	Yes (combo at GP)
20	57.3%	28.6%	57.3%	Yes (combo at GP)
30	22.1%	17.8%	22.1%	Yes (both near GP)
40	3.2%	3.2%	3.2%	Yes (both at GP)

The constraint becomes active for the combo arm at approximately year 8–9, when the unconstrained log-normal OS exceeds the GP survival. The pembro arm, with its shorter survival, does not reach the GP floor until approximately year 30. By year 40, both arms converge to the GP floor (3.2%).

4.3 OS versus General-Population Survival



The GP-constrained curves (solid lines) are capped at the general population survival floor from month 60 onward. The unconstrained log-normal extrapolation (dotted lines) produces implausibly high 40-year survival for the combo arm (76.4%).

Figure S4: Overall survival versus general population. Solid lines: GP-constrained model (base case). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US general population survival from age 61 (64% male-weighted). The vertical dotted line at 5 years marks the onset of the GP mortality constraint. The unconstrained combo arm (76.4% at 40 years) is clearly implausible compared to the GP floor (0% at age 101).

4.4 Structural Quality Assurance

The model was run for 480 monthly cycles (40 years) with structural checks at every cycle. The results are summarized in Table S6.

Table S6: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0
State-sum convergence (all cycles)	$1.0000 \pm < 10^{-12}$

The model satisfies all structural constraints at every cycle, confirming the robustness of the ratio-constrained PSM approach. The $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ ordering is maintained without any clamping, and all health-state probabilities remain non-negative and sum to unity.

5 External Validation

External validation assessed the plausibility of the pembrolizumab arm survival predictions against published landmark data from two independent adjuvant melanoma trials: KEYNOTE-054 (pembrolizumab vs. placebo, stage III, $N = 1,019$) [6] and CheckMate 238 (nivolumab vs. ipilimumab, stage IIIB–IV, $N = 906$) [7]. These validation targets represent populations that differ from KEYNOTE-942 in stage distribution, agent, and follow-up duration; the comparison is therefore a qualitative plausibility check rather than a formal calibration target.

5.1 External Validation Landmarks

Table S7: External Validation of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	49.1%	~50% (nivo)
RFS	7	31.9%	50.0% (46–55%)	~44% (nivo)
DMFS	5	53.8%	~58%	58% (nivo)
DMFS	7	44.9%	54% (50–59%)	~50% (nivo)
OS	5	74.5%	NR	76% (nivo)
OS	7	64.2%	NR	~70% (nivo)
RFS	10	23.2%	~47% (extrap.)	~40% (nivo, est.)
DMFS	10	35.0%	~50% (extrap.)	~45% (nivo, est.)
OS	10	51.9%	NR	~65% (nivo, est.)

NR: Not reported (OS data immature or crossover-contaminated in KEYNOTE-054). nivo: CheckMate 238 evaluated nivolumab, not pembrolizumab. Extrap. = extrapolated, est. = estimated.

Interpretation: The model predicts lower RFS and DMFS for the pembrolizumab arm compared to KEYNOTE-054 and CheckMate 238 at all time points. This is expected because KEYNOTE-942 enrolled a higher-risk population (100% stage IIIB–IV with high-risk features, including mucosal and acral melanoma subtypes) compared to KEYNOTE-054 (stage IIIA–C,

predominantly low-risk) and CheckMate 238 (stage IIIB–IV, but with fewer high-risk subtypes). The OS predictions are consistent with CheckMate 238 (5-year OS: 74.5% vs. 76%, 7-year OS: 64.2% vs. $\sim 70\%$), suggesting that the long-term survival extrapolation is clinically plausible despite the higher-risk population.

5.2 Longitudinal External Validation

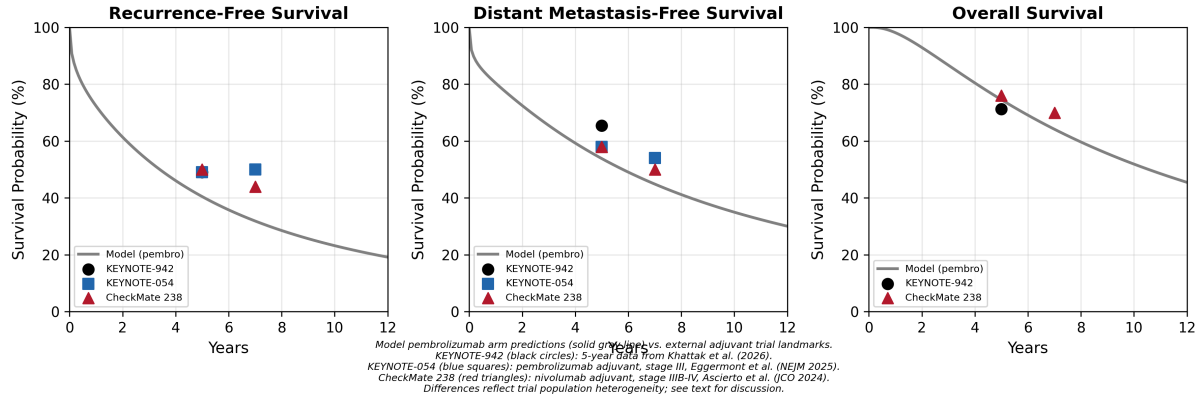


Figure S5: External validation of pembrolizumab arm predictions. Gray solid line: model-predicted survival for the pembrolizumab arm. Black circles: KEYNOTE-942 observed 5-year data. Blue squares: KEYNOTE-054 landmark data. Red triangles: CheckMate 238 landmark data. The model's lower RFS/DMFS predictions reflect the higher-risk KEYNOTE-942 population. The OS predictions are consistent with CheckMate 238.

6 Model Inputs

6.1 Complete Parameter Table

Table S8: Full Model Parameter Inventory

Parameter	Base	DSA Low	DSA High	PSA Distribution
<i>Clinical Parameters</i>				
OS μ – combo	4.428	4.228	4.628	Normal (SE from 0.000)
OS σ – combo	0.478	0.378	0.578	Normal (SE from 0.000)
OS μ – pembro	3.725	3.525	3.925	Normal (SE from 0.000)
OS σ – pembro	0.574	0.474	0.674	Normal (SE from 0.000)
r _{DM} μ – combo	4.107	3.907	4.307	Normal (SE from 0.000)
r _{DM} σ – combo	0.432	0.332	0.532	Normal (SE from 0.000)
r _{DM} μ – pembro	2.849	2.649	3.049	Normal (SE from 0.000)
r _{DM} σ – pembro	0.934	0.834	1.034	Normal (SE from 0.000)
r _{LR} μ – combo	3.367	3.167	3.567	Normal (SE from 0.000)
r _{LR} σ – combo	0.624	0.524	0.724	Normal (SE from 0.000)
r _{LR} μ – pembro	2.750	2.550	2.950	Normal (SE from 0.000)
r _{LR} σ – pembro	0.638	0.538	0.738	Normal (SE from 0.000)
Treatment duration (pembro)	18 cycles (2 yr)	9	24	Fixed
Treatment duration (intismeran)	9 doses (1 yr)	6	12	Fixed
Baseline age	61 years	55	67	Fixed
Male proportion	64%	–	–	Fixed
<i>Costs (2026 USD)</i>				
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Gamma (CV=0.5)
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.5)
Tumor sequencing	\$1,000	\$500	\$2,000	Gamma (CV=0.5)
Administration (per cycle)	\$500	\$250	\$750	Gamma (CV=0.5)
AE management (one-time)	\$15,000	\$7,500	\$22,500	Gamma (CV=0.5)
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.5)
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.5)
<i>Utilities</i>				
RF state	0.83	0.80	0.86	Beta (SE=0.03)
LR state	0.64	0.61	0.67	Beta (SE=0.03)
DM state	0.55	0.52	0.58	Beta (SE=0.03)
AE disutility (one-time)	0.05	0.00	0.10	Beta (SE=0.02)
<i>Economic Parameters</i>				
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)
Time horizon	40 years	10	20	Fixed (scenario)
Willingness-to-pay threshold	\$100,000–\$150,000/QALY	\$50,000	\$200,000	Fixed (scenario)
Cost base year	2026 USD	–	–	–
GP mortality source	US 2019 Life Tables	–	–	–

Abbreviations: DSA = deterministic sensitivity analysis; PSA = probabilistic sensitivity analysis; SE = standard error; CV = coefficient of variation; WAC = wholesale acquisition cost; CMS = Centers for Medicare & Medicaid Services; NVSS = National Vital Statistics System. Survival parameters were fitted via nonlinear least squares; Jacobian-derived SEs were used for PSA. Costs were inflated to 2026 USD using the medical care component of the Consumer Price

Index (CPI).

7 Additional Results

7.1 Cost Decomposition

Table S9: Cost Decomposition by Component

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration/AE	\$15,493	\$493	\$15,000
LR management	\$83,196	\$69,500	\$13,696
DM management	\$342,542	\$425,276	-\$82,734
Total	\$860,118	\$713,157	\$146,961

7.2 QALY Decomposition

Table S10: QALY Decomposition by Health State

State	Combo	Pembro	Incremental
RF	8.32	4.49	3.83
LR	1.48	1.24	0.24
DM	1.31	1.62	-0.32
Total	11.10	7.35	3.76

8 Probabilistic Sensitivity Analysis

8.1 PSA Distributions and Covariance

Table S11: PSA Parameter Distributions

Parameter	Distribution	Mean	SE
OS mu – combo	Normal	4.40	0.35
OS sigma – combo	Normal	0.48	0.10
OS mu – pembro	Normal	3.75	0.30
OS sigma – pembro	Normal	0.55	0.12
Utility RF	Beta	0.83	0.04
Utility LR	Beta	0.64	0.04
Utility DM	Beta	0.55	0.04
Cost LR monthly	Gamma	3000	600
Cost DM monthly	Gamma	12000	2400

8.2 PSA Convergence

[Figure S4: PSA convergence – mean ICER at 100/500/1000/5000 iterations]

8.3 Cost-Effectiveness Plane

[Figure S5: Cost-effectiveness plane (1,000 PSA iterations)]

9 Structural Scenario Analyses

9.1 Full Scenario Results

Table S12: Structural Scenario Analysis Results

Scenario	ΔCost	ΔQALY	ICER	NMB @\$150K
Base case	\$146,961	3.76	\$39,105	\$417,039
Unconstrained	\$1,122,111	14.90	\$75,311	\$1,112,889
Waning (5–10y convergence)	\$105,360	1.83	\$57,691	\$168,144
No direct OS benefit	-\$29,379	0.64	Dominant	\$125,379
Weibull OS	–	–	–	–
10-year horizon	–	–	–	–
20-year horizon	–	–	–	–

10 CHEERS 2022 Checklist

Table S13: CHEERS 2022 Checklist

No.	Item	Reported Location	
1	Title	Yes	Title page
2	Abstract	Yes	Abstract
3	Background	Yes	Introduction
4	Health economic analysis plan	Yes	Methods
5	Model	Yes	Methods, Appendix A
6	Population	Yes	Methods
7	Setting	Yes	Methods
8	Comparators	Yes	Methods
9	Perspective	Yes	Methods
10	Time horizon	Yes	Methods
11	Discount rate	Yes	Methods
12	Outcomes	Yes	Methods
13	Health states	Yes	Appendix A
14	Measurement of outcomes	Yes	Methods
15	Resource use	Yes	Methods
16	Currency	Yes	Methods
17	Model choice	Yes	Methods
18	Assumptions	Yes	Methods
19	Analytic methods	Yes	Methods
20	Study parameters	Yes	Methods, Appendix D
21	Incremental results	Yes	Results
22	Uncertainty	Yes	PSA, DSA
23	Heterogeneity	Yes	Subgroup
24	Discussion	Yes	Discussion
25	Limitations	Yes	Discussion
26	Funding	Yes	Declarations
27	Conflicts	Yes	Declarations
28	Reproducibility	Yes	Appendix

References

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